Student ID:

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 2 **Winter Semester 2024-2025**

DURATION: 30 MIN **FULL MARKS: 20**

CSE 4107: Structured Programming I

Figures in the right margin indicate marks.

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1. Find the output of the following program: **6 (CO1) (PO2)**

```
#include <stdio.h>
int main(){
    int i,j;
    for(i=0,j=10;i<j;i++,j--)
    {
        if(j%2==0){
            printf("%d %d\n",i,j);
        }
    }

    return 0;
}
```

2. Write a C program that takes a positive integer, `n`, from the user and prints a triangle using the '`*`' character, where the size of the triangle is determined by `n`. A sample input and output pair is shown below: **8 (CO3) (PO1)**

```
Enter n: 5
    *
   ***
  *****
 *******
*********
```

3. Explain the differences in the execution of the statements between the following two code snippets: **6 (CO2) (PO1)**

```
if (condition)
    statement1;
else
    statement2;
```

Code Snippet 1

```
if (condition) {
    statement1;
}
statement2;
```

Code Snippet 2